

DATABASE DESIGN DOCUMENT

VIDYA SAATHI

AI-Powered Personalized Learning Companion for Students

Team VidyaSetu

1. Introduction

This document defines the database design for Vidya Saathi, derived from the six Phase 1 modules of the SRS: Registration & Profile Management, Subject & Topic Selection, Doubt Resolution, Quiz Generation & Assessment, Adaptive Learning Path Management, and Language Localization Management.

Design methodology follows the entity-first approach: entities and attributes are listed from the SRS, relationships between entity pairs are identified, and tables are generated from those relationships — never created directly. A 1:M relationship generates two tables (the second carrying a foreign key to the first). An M:M relationship generates three tables, with a junction/child table holding both primary keys.

2. List of Entities

1. Registration & Profile Management

- Class
- Board
- Language
- Security_Question
- Account_Status
- Student

2. Subject & Topic Selection

- Subject
- Topic
- Student_Topic_Selection

3. Doubt Resolution (AI Companion)

- Doubt

4. Quiz Generation & Assessment

- Quiz
- Question
- Quiz_Attempt

5. Adaptive Learning Path Management

- Learner_Profile
- Recommendation_Rule

6. Language Localization Management

- User_Language_Preference

Note: Language (Module 1) and Student (Module 1) are reused here — no new master entity is created.

3. List of Attributes

1. Registration & Profile Management

- **Class Provisioning**
 - Class_id (PK)
 - Class_name
- **Board Provisioning**
 - Board_id (PK)
 - Board_name
- **Language Provisioning**
 - Language_id (PK)
 - Language_name
 - Language_code
- **Security Question Provisioning**
 - Security_ques_id (PK)
 - Question_text
- **Account Status Provisioning**
 - Account_status_id (PK)
 - Status_name
 - Description
- **Student Provisioning**
 - Student_id (PK)
 - First_name
 - Last_name
 - Gender
 - Date_of_Birth
 - Username
 - Password
 - Email
 - Phone_number
 - Security_answer
 - FK Class_id, FK Board_id, FK Language_id, FK Security_ques_id, FK Account_status_id

2. Subject & Topic Selection

- **Subject Provisioning**
 - Subject_id (PK)
 - Subject_name

- **Topic Provisioning**
 - Topic_id (PK)
 - Topic_name
 - Class_applicability
 - FK Subject_id
- **Student_Topic_Selection Provisioning**
 - Selection_id (PK)
 - Last_selected_timestamp
 - Selection_count
 - FK Student_id, FK Topic_id

3. Doubt Resolution (AI Companion)

- **Doubt Provisioning**
 - Doubt_id (PK)
 - Question_text
 - AI_answer_text
 - Language
 - Timestamp
 - Helpful_rating
 - FK Student_id, FK Subject_id, FK Topic_id

4. Quiz Generation & Assessment

- **Quiz Provisioning**
 - Quiz_id (PK)
 - Difficulty_level
 - FK Topic_id
- **Question Provisioning**
 - Question_id (PK)
 - Question_text
 - Option_A – Option_D
 - Correct_option
 - Explanation_text
 - FK Quiz_id
- **Quiz_Attempt Provisioning**
 - Attempt_id (PK)
 - Score
 - Time_taken
 - Date

- FK Student_id, FK Quiz_id

5. Adaptive Learning Path Management

- **Learner_Profile Provisioning**
 - Learner_profile_id (PK)
 - Mastery_score
 - Attempts_count
 - Last_updated_timestamp
 - FK Student_id, FK Topic_id
- **Recommendation_Rule Provisioning**
 - Rule_id (PK)
 - Trigger_condition
 - Recommended_action

6. Language Localization Management

- **User_Language_Preference Provisioning**
 - Preference_id (PK)
 - FK Student_id, FK Language_id

4. Relationships Between Entities

1. Registration & Profile Management

Class —————▶ Student (1 to M)
Board —————▶ Student (1 to M)
Language —————▶ Student (1 to M)
Security_Question —————▶ Student (1 to M)
Account_Status —————▶ Student (1 to M)

2. Subject & Topic Selection

Subject —————▶ Topic (1 to M)
Student —————▶ Topic (M to M (Student_Topic_Selection))

3. Doubt Resolution (AI Companion)

Student —————▶ Doubt (1 to M)
Subject —————▶ Doubt (1 to M)
Topic —————▶ Doubt (1 to M)

4. Quiz Generation & Assessment

Topic —————▶ **Quiz** (1 to M)

Quiz —————▶ **Question** (1 to M)

Student —————▶ **Quiz** (M to M (Quiz_Attempt))

5. Adaptive Learning Path Management

Student —————▶ **Topic** (M to M (Learner_Profile))

Recommendation_Rule is a system business-rule lookup evaluated by application logic against Learner_Profile records; it is not tied by a physical foreign key in Phase 1.

6. Language Localization Management

Language —————▶ **Student** (M to M (User_Language_Preference))

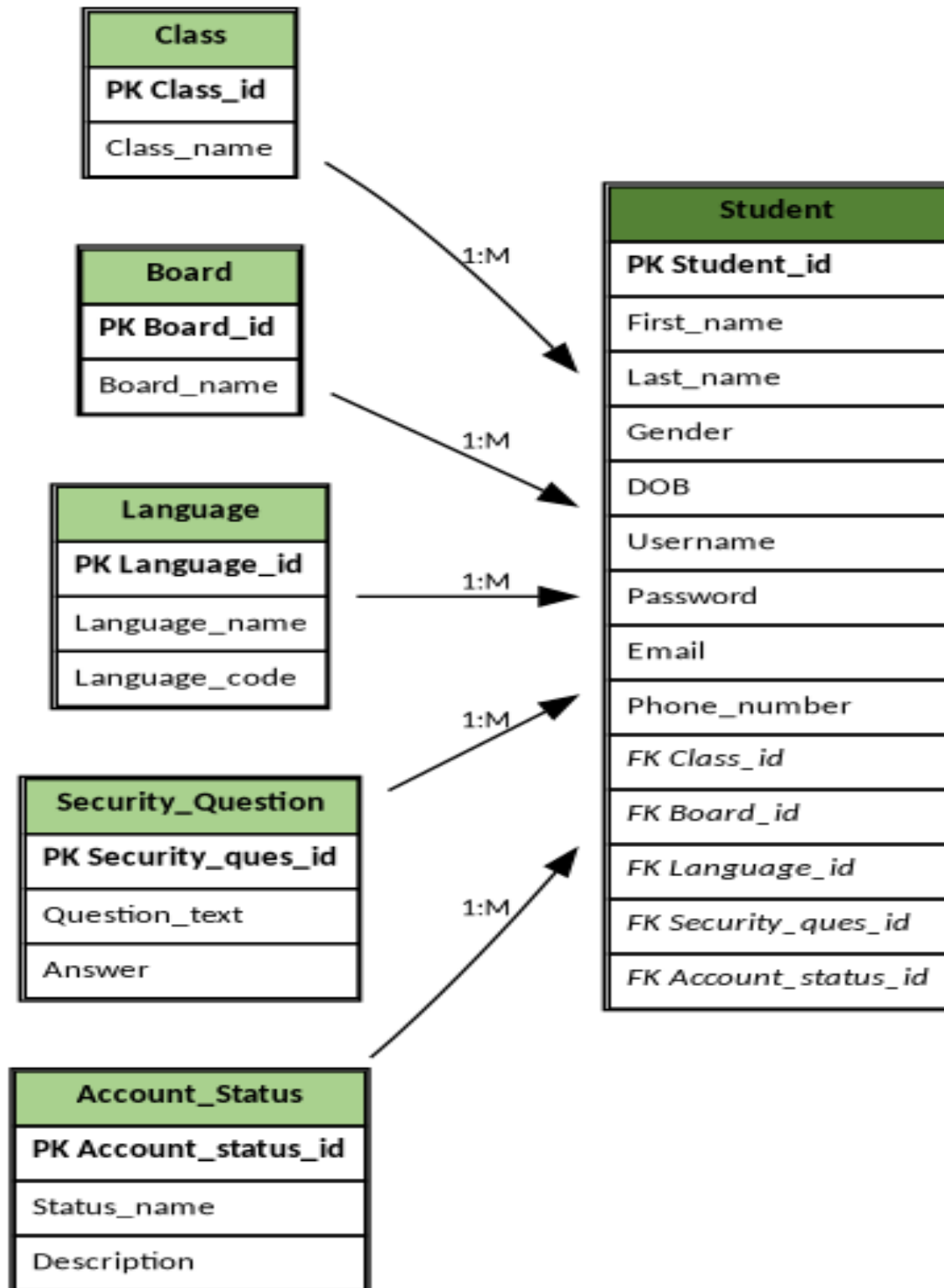
Note: 1 to M (One to Many).

M to M (Many to Many) — junction/child table named in brackets.

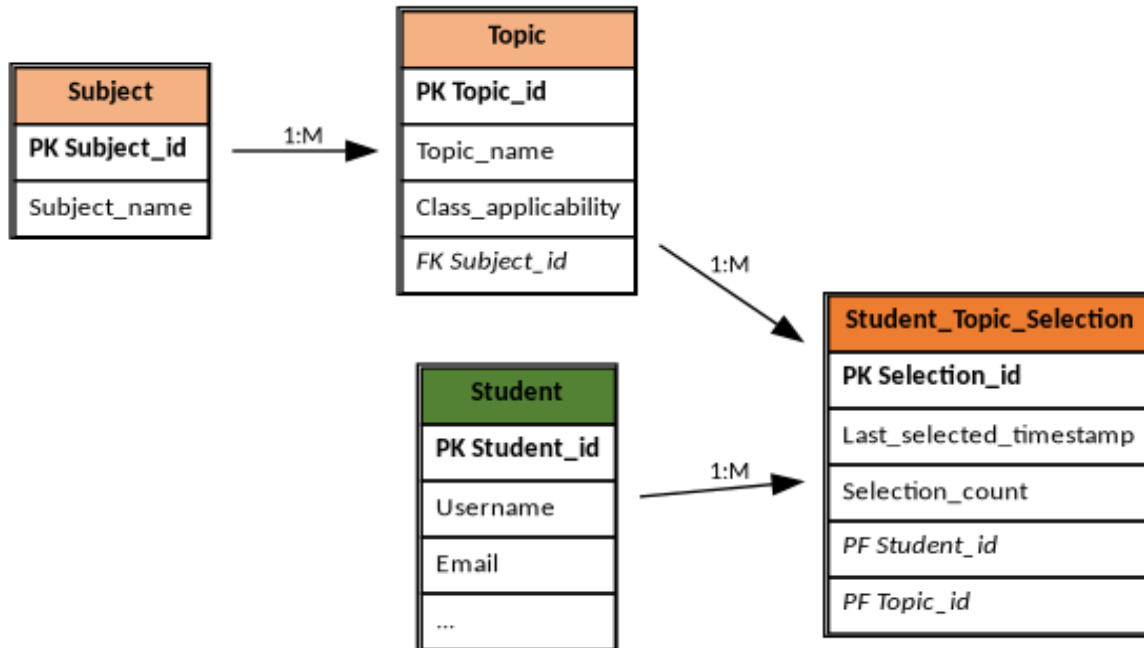
“—————▶” represents the relationship direction between two entities.

5. Database Design (ER Diagrams)

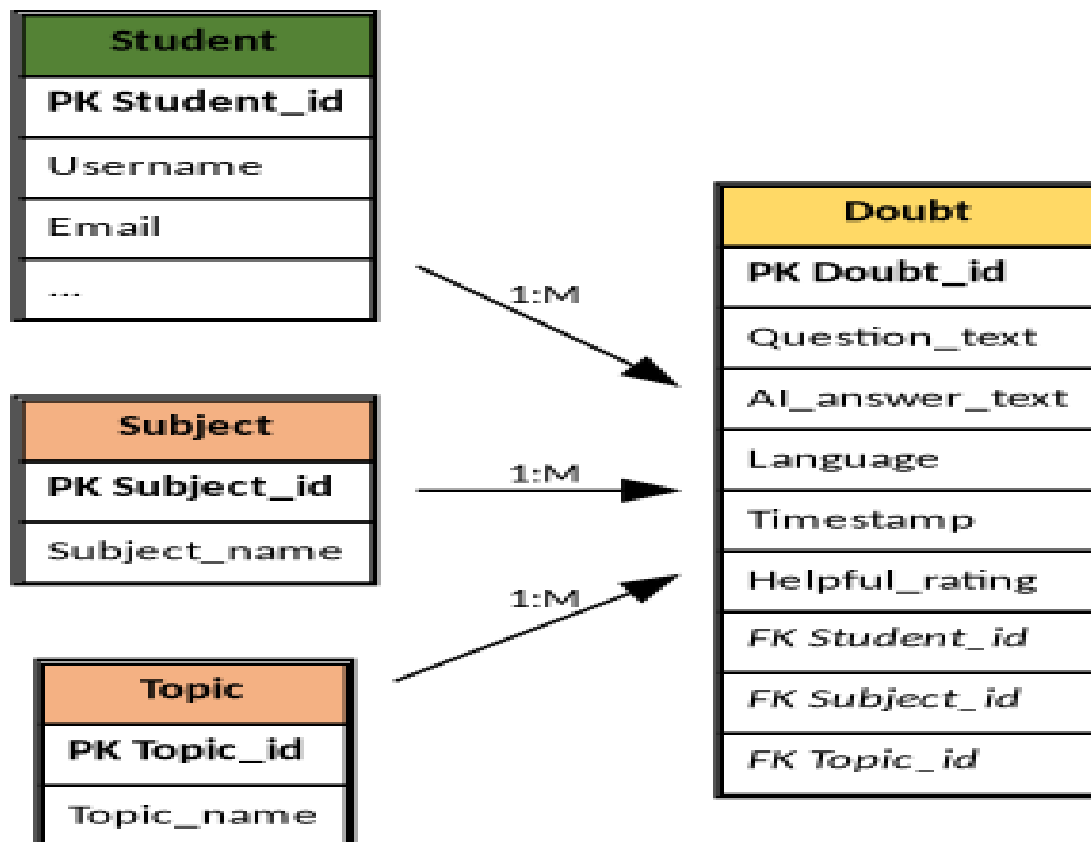
1. Registration & Profile Management



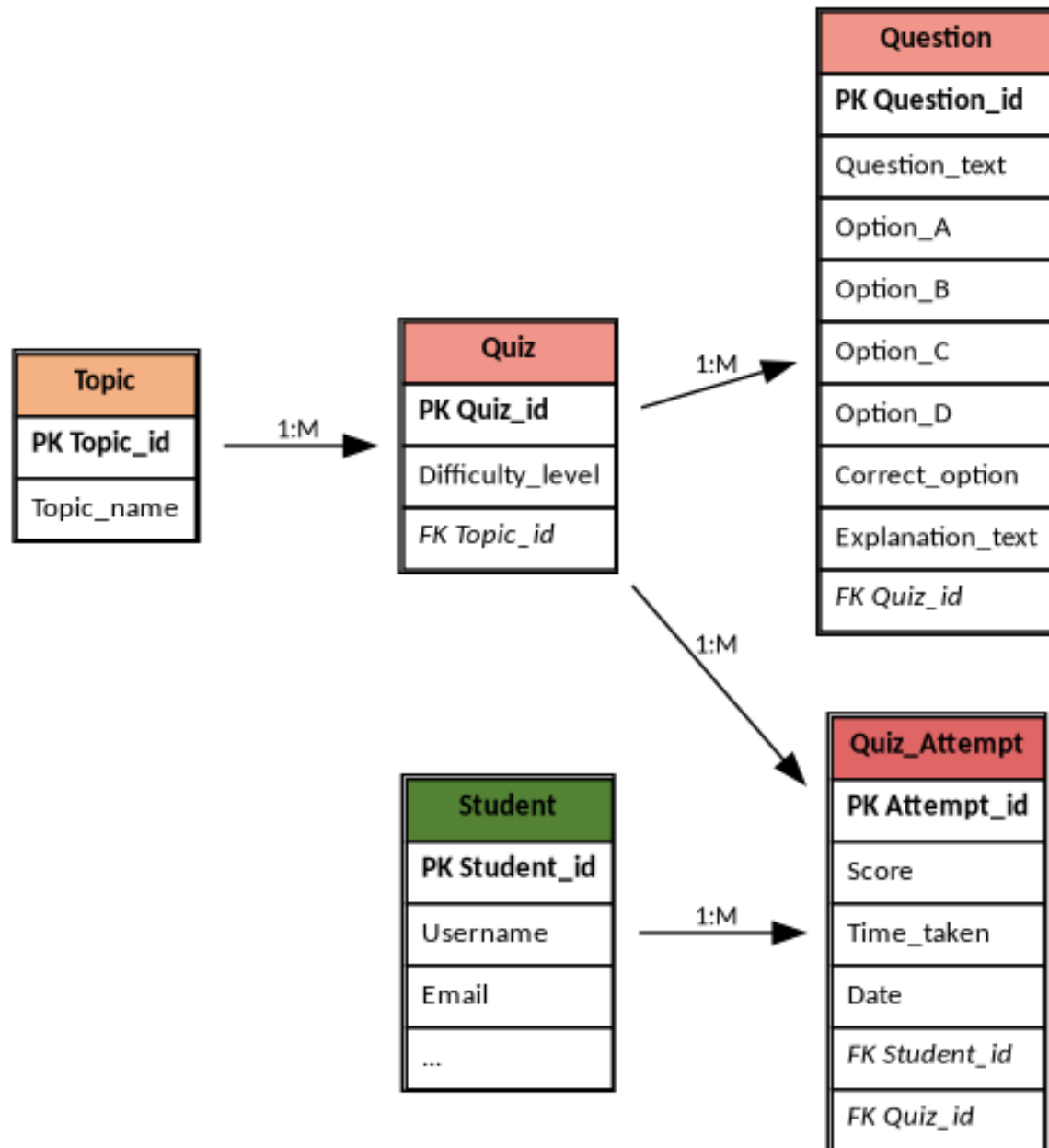
2. Subject & Topic Selection



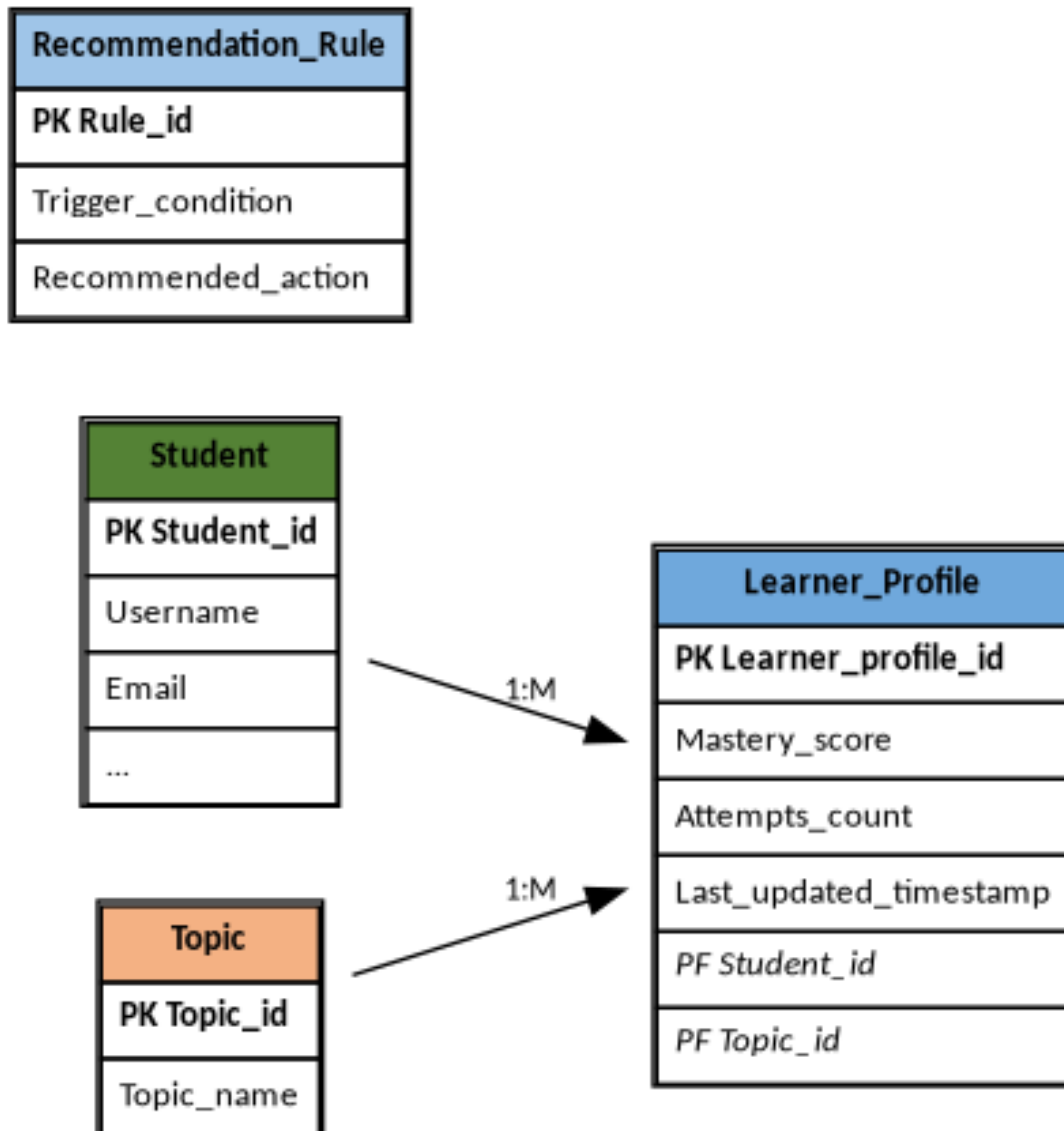
3. Doubt Resolution (AI Companion)



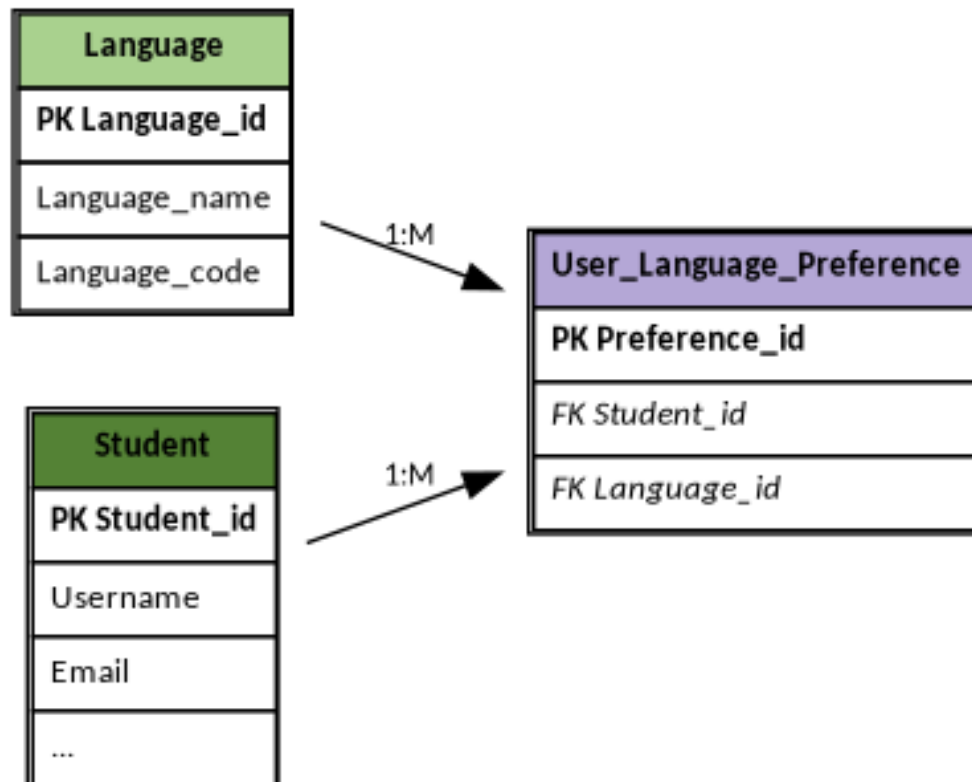
4. Quiz Generation & Assessment



5. Adaptive Learning Path Management



6. Language Localization Management



6. Generating Tables

Tables generated from the entities and relationships above. Every FK column references the PK of its parent table.

1. Registration & Profile Management

Table: Class		
Column Name	Data Type	Key / Notes
Class_id	INT	PK
Class_name	VARCHAR(20)	—

Table: Board		
Column Name	Data Type	Key / Notes
Board_id	INT	PK
Board_name	VARCHAR(50)	—

Table: Language		
Column Name	Data Type	Key / Notes
Language_id	INT	PK
Language_name	VARCHAR(30)	—
Language_code	VARCHAR(5)	—

Table: Security_Question		
Column Name	Data Type	Key / Notes
Security_ques_id	INT	PK
Question_text	VARCHAR(255)	—

Table: Account_Status		
Column Name	Data Type	Key / Notes
Account_status_id	INT	PK
Status_name	VARCHAR(20)	—
Description	VARCHAR(255)	—

Table: Student		
Column Name	Data Type	Key / Notes
Student_id	INT	PK
First_name	VARCHAR(50)	—
Last_name	VARCHAR(50)	—
Gender	VARCHAR(10)	—
Date_of_Birth	DATE	—
Username	VARCHAR(50)	—
Password	VARCHAR(255)	hashed
Email	VARCHAR(100)	—
Phone_number	VARCHAR(15)	—
Security_answer	VARCHAR(255)	—
Class_id	INT	FK → Class
Board_id	INT	FK → Board
Language_id	INT	FK → Language
Security_ques_id	INT	FK → Security_Question
Account_status_id	INT	FK → Account_Status

2. Subject & Topic Selection

Table: Subject		
Column Name	Data Type	Key / Notes
Subject_id	INT	PK
Subject_name	VARCHAR(50)	—

Table: Topic		
Column Name	Data Type	Key / Notes
Topic_id	INT	PK
Topic_name	VARCHAR(100)	—
Class_applicability	VARCHAR(20)	—
Subject_id	INT	FK → Subject

Table: Student_Topic_Selection		
Column Name	Data Type	Key / Notes
Selection_id	INT	PK
Last_selected_timestamp	DATETIME	—
Selection_count	INT	—
Student_id	INT	FK → Student
Topic_id	INT	FK → Topic

3. Doubt Resolution (AI Companion)

Table: Doubt		
Column Name	Data Type	Key / Notes
Doubt_id	INT	PK
Question_text	TEXT	—
AI_answer_text	TEXT	—
Language	VARCHAR(20)	—
Timestamp	DATETIME	—
Helpful_rating	VARCHAR(10)	—
Student_id	INT	FK → Student
Subject_id	INT	FK → Subject
Topic_id	INT	FK → Topic

4. Quiz Generation & Assessment

Table: Quiz		
Column Name	Data Type	Key / Notes
Quiz_id	INT	PK
Difficulty_level	VARCHAR(20)	—
Topic_id	INT	FK → Topic

Table: Question		
Column Name	Data Type	Key / Notes
Question_id	INT	PK
Question_text	TEXT	—
Option_A	VARCHAR(255)	—
Option_B	VARCHAR(255)	—
Option_C	VARCHAR(255)	—
Option_D	VARCHAR(255)	—
Correct_option	VARCHAR(5)	—
Explanation_text	TEXT	—
Quiz_id	INT	FK → Quiz

Table: Quiz_Attempt		
Column Name	Data Type	Key / Notes
Attempt_id	INT	PK
Score	DECIMAL(5,2)	—
Time_taken	INT	seconds
Date	DATETIME	—
Student_id	INT	FK → Student
Quiz_id	INT	FK → Quiz

5. Adaptive Learning Path Management

Table: Learner_Profile		
Column Name	Data Type	Key / Notes
Learner_profile_id	INT	PK
Mastery_score	DECIMAL(5,2)	—
Attempts_count	INT	—
Last_updated_timestamp	DATETIME	—
Student_id	INT	FK → Student
Topic_id	INT	FK → Topic

Table: Recommendation_Rule		
Column Name	Data Type	Key / Notes
Rule_id	INT	PK
Trigger_condition	VARCHAR(255)	—
Recommended_action	VARCHAR(255)	—

6. Language Localization Management

Table: User_Language_Preference		
Column Name	Data Type	Key / Notes
Preference_id	INT	PK
Student_id	INT	FK → Student
Language_id	INT	FK → Language